

AHMAD HAZWAN SYAUQEE BIN HUSAIRI

Computer Science Graduate | Mobile & Full-Stack Development | AI & Cloud Systems

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SUMMARY

Computer Science (Intelligent Computing) graduate from Universiti Sains Malaysia with hands-on industrial experience at **MIMOS Berhad**, where I modernised a legacy inventory system end-to-end. My final year project applies **YOLOv8** to real-time road hazard detection, backed by a full-stack cloud deployment (React PWA, Flask on Cloud Run, Firebase) and a human-in-the-loop MLOps retraining pipeline. Most recently built a Flutter product catalog app with pagination, debounced search, and explicit loading, error, and empty states. Comfortable moving across mobile, computer vision, IoT hardware, and cloud-native web systems, and looking to bring that range to a mobile or software engineering team.

EDUCATION

Bachelor of Computer Science (Intelligent Computing) with Honours | *Universiti Sains Malaysia, Penang* 2022 – 2026

- Final CGPA: 3.30

Foundation in Engineering | *Pusat Asasi UiTM Dengkil, Selangor* 2021 – 2022

- GPA: 4.00 | MUET: Band 4.5

Sijil Pelajaran Malaysia (SPM) | *SMK Seksyen 4 Kota Damansara, Selangor* 2016 – 2020

- Results: 5A 4B

PROFESSIONAL EXPERIENCE

Industrial Intern | *MIMOS Berhad* Mar 2025 – Sep 2025

- Extended a legacy **PHP/MySQL** inventory system (DCIMS) by designing and building two new modules, End User Computing and Software asset management, from scratch, including new database schemas, audit trails, and assignment history tracking.
- Prototyped workflows in **Figma** and built the front-end for device listings, dashboards, and data entry forms with JavaScript/jQuery and Bootstrap, adding real-time feedback via toast notifications and a recent-activity dashboard.
- Implemented role-based access control across five user tiers (Server, Network, EUC, Software, and Super Admin) and restructured device categorisation from a location-based to a category-based system, improving lookup and usability across 650+ device records.
- Authored UAT test scripts and led acceptance testing sessions with admin groups across four modules, resolving permission errors, data validation gaps, and usability issues raised by end users.
- Migrated the system from a local **XAMPP** environment to a live testing server via WinSCP, debugging permission and configuration issues to stabilise deployment.
- Produced an operations user manual and system architecture diagram and worked cross-functionally with senior engineers to align features with end-user operational needs.

TECHNICAL PROJECTS

Road Hazard Detection System | *Final Year Project* 2025 – 2026

- Built an end-to-end computer vision pipeline using **YOLOv8**, reaching 83.28% mAP@0.5 on road crack and pothole detection.
- Shipped a citizen-facing React PWA, an admin dashboard, and a contractor portal, backed by a Flask API on **Google Cloud Run** and Firebase Firestore/Auth.
- Designed a human-in-the-loop (HITL) MLOps pipeline for continuous model retraining from field-reported data.
- Presented at PIXEL 2026, winning **Silver**; entered into the IEM IDEAS Competition 2026.

Product Catalog App | *Flutter* | github.com/HazzyWan/product-catalog-app 2026

- Built a **Flutter** app consuming the DummyJSON REST API, with infinite-scroll pagination, a product detail screen, and debounced server-side search.
- Modelled screen state as a **sealed class** (loading, empty, error with retry, success) and split the code into data and presentation layers, with a repository handling pagination.
- Added pull-to-refresh, image loading and error placeholders, and **unit tests** for JSON parsing.

Smart Drain: IoT Blockage & Flood Detection System 2025

- Engineered a smart-city IoT hardware prototype using an **ESP32** microcontroller and ultrasonic sensors to monitor drain water levels.
- Integrated hardware telemetry with a cloud backend via **MQTT**, Firebase, and Google Cloud Platform for real-time remote alerts.

Business Performance Analytics Dashboard 2026

- Built an interactive Tableau dashboard analysing the Superstore retail dataset to expose the gap between revenue and net margin across categories, regions, and time.
- Identified a high-volume, low-margin pattern in the Furniture category using side-by-side bar charts, dual-axis trend lines, and treemaps, and translated findings into three actionable business recommendations.

Toxland VR: Hop & Help | *Animation & VR Coursework* 2026

- Migrated an existing VR board game to Unity 6.3 for the **PICO 4** headset, resolving compatibility issues across the upgrade.
- Produced the accompanying project report and presentation materials.

SKILLS

Technical: C++, Python, Java, PHP, SQL, R, JavaScript, HTML/CSS, Node.js, React.js, Vite

Mobile: Flutter, Dart, Provider, React PWA

Tools: Git & GitHub, VS Code, Android Studio / Emulator
Cloud & Systems: Google Cloud Platform (Cloud Run), Firebase, MQTT, YOLOv8, IoT (ESP32)
Data & Analytics: Oracle APEX, sentiment analysis, Altair AI Studio, Tableau
Game & VR: Unity (6.3), PICO 4 VR development
Software: Canva, Blender, Adobe Photoshop, Figma, Microsoft Office
Languages: Malay (native), English (fluent)

COMPETITIONS & EXTRACURRICULARS

Silver Award, PIXEL 2026 <i>School of Computer Sciences, Universiti Sains Malaysia</i>	2026
• Awarded Silver for the Road Hazard Detection System at PIXEL 2026, USM's School of Computer Sciences project showcase.	
Entrant, IEM IDEAS Competition 2026 <i>Institution of Engineers Malaysia (IEM), Individual Category</i>	2026
• Entered the Road Hazard Detection System into IEM's Innovation, Design, and Engineering for Advanced Sustainability (IDEAS) Competition.	
Participant, Data Challenge 4.0 <i>UKM Statistics Club</i>	2025
• Completed a data analytics competition run by Universiti Kebangsaan Malaysia's Statistics Club, Feb–May 2025.	

CERTIFICATIONS

- Certificate of Industrial Internship, MIMOS Berhad — Sept 2025 (24-week residency in full-lifecycle software engineering and legacy modernisation)
- Data-Driven Decision Making: AIoT and V-ONE Data Analytics Perspective, ViTrox Academy — Dec 2025

REFERENCE

Dr. Nurul Hashimah — Final Year Project Supervisor, Universiti Sains Malaysia
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